GROUP 5 DISCUSSION REPORT

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INTRODUCTION

Supervised learning - training models on labeled data to predict outcomes, e.g., logistic regression.

steps/process of building supervised learning models

->Data preparation-cleaning and encoding features.

->Model Training-fitting the model to labeled data.

->Evaluation-accessing model performance using metrics like accuracy, precision, recall,

and F1-score.

->Prediction and Deployment- once validated, the model predicts outcomes for new data.

Real-world applications include predicting customer churn and medical diagnosis.

Key Terms and concepts:

1. Labeled data - A dataset where each input is paired with a corresponding output.
2. Features(X) - The input variables or attributes used to make predictions.
3. Target(y) - The output variable/label that the model is trying to predict.
4. Training Data - The portion of the dataset used to teach the model(both input variables and their corresponding labels).
5. Testing Data - Separate portion of the dataset used to evaluate the model's performance on unseen data.
6. Loss Function - Mathematical function that measures the difference between the model's predictions and the actual labels.
7.

Tasks in Supervised Learning:

1. Classification - predicting discrete labels eg whether an email is spam or not spam.
2. Regression - predicting continuous values eg forecasting house prices.
3.

INTRODUCTION TO LOGISTIC REGRESSION:

Logistic regression is a fundamental algorithm in supervised learning used for classification tasks; it predicts probabilities for discrete outcomes.

Maximum Likelihood Estimation(MLE): In logistic regression, the model tries to find the best parameters(weights and bias) that make the predictions match the actual data as closely as possible.

Gradient Descent: To find the best parameters, the model uses an optimization technique called gradient descent.

Key Terms and Concepts

1. Sigmoid Function: A mathematical function that maps any real-valued number to a value between 0 and 1.
2. Log-Odds: Logistic regression models the log-odds (logarithm of the odds) of the dependent variable being 1 as a linear function of the independent variables.
3. Binary Classification: Logistic regression is commonly used for tasks with two possible outcomes (e.g., spam/not spam, churn/no churn).
4. Multiclass Classification: Logistic regression can also handle multi-class problems using techniques like one-vs-rest (OvR) or multinomial logistic regression.
5. Maximum Likelihood Estimation (MLE): A method used to estimate the model parameters by maximizing the likelihood of observing the given data.
6.

How Does Logistic Regression Work?

Significance in Data Science

->Interpretability: Logistic regression provides interpretable coefficients, indicating the direction and magnitude of each feature's impact.

->Efficiency: It is computationally efficient for both binary and multiclass classification tasks.

->Baseline Model: Logistic regression is often used as a baseline model due to its simplicity and robustness.

Process in Logistic Regression

1. Define the Problem and Goals:

2. Data Preparation and Exploration: Gather, clean, and preprocess the data.

3. Model Specification and Design:

4. Model Training and Parameter Estimation: Use the training data to fit the model.

5. Model Evaluation and Diagnostics: Evaluate the model's performance on test data.

6. Model Interpretation and Communication: Interpret the results of the logistic regression.

7. Refinement and Deployment:

CONCLUSION

Completed learning content on supervised learning and Logistic Regression.

REFERENCES

1. Canvas
2. https://www.youtube.com/watch?v=yIYKR4sgzI8